

IIT Foundation Module for Class VII

Physics

- Theme 1: Matter
 1. Kinetic Theory of Matter
 - Three states of matter in terms of movement of particles
 - Energy content in the three states of matter
 2. Change of state in matter using the Kinetic theory
 - Boiling
 - Vaporization
 - Melting
 - Fusion
 - Evaporation
 - Condensation
 - Sublimation
 - Deposition
 - Freezing
 3. Change of state diagrams (using the terms mentioned above)

- Theme 2: Physical Quantities and Measurement
 1. Measurement of Density of irregular solids using
 - Eureka Can
 - Measuring Cylinder
 2. Measurement of Density of Fluids
 - Basic Concept
 - Concept of Floatation and sinking of a substance (relate to density)
 - Comparison of densities in the three states of matter

- Theme 3: Force and Pressure
 1. Turning effect of force (moment of force): concept, definition and calculation

2. Pressure

- Definition
- Unit
- Calculation of pressure in simple cases
- Pressure exerted by liquids (Qualitative only)
- Pressure exerted by gases – Atmospheric pressure (Qualitative only)

• Theme 4: Energy

1. Concept of Work

- Unit of Work (Joule)
- Calculation of Work done in simple cases

2. Kinetic Energy – Basic Concept

3. Potential Energy – Basic Concept

4. Gravitational Potential Energy

- Calculation of kinetic and potential energies from a set of given data (Simple problems and assuming $g=10 \text{ m/s}^2$)

5. Energy transformation in common daily life situations

6. Difference between Energy and Power

• Theme 5: Light Energy

1. Refraction

- Definition
- Examples of Refraction

2. Curved Mirrors

- Convex
- Concave
- Reflecting surface (Convex and Concave)
- Uses of Curved mirrors
- Terms related to Curved mirrors – Focus, Principal Axis, centre of curvature, radius of curvature
- Rules for making ray diagrams of Spherical mirrors
- Real and Virtual Images

- Ray diagrams with curved mirrors where real images are formed
- 3. Dispersion of white light into constituent colours

- Theme 6: Heat Transfer

1. Difference between Boiling and Evaporation
2. Thermal Expansion
 - Linear Expansion
 - Volume Expansion
 - Superficial Expansion
 - Compare expansivity in Solids, Liquids and Gases
 - Examples and Real-world applications

- Theme 7: Sound

1. Pitch and Frequency
2. Pitch and frequency in relation to working of musical instruments (Wind, membrane and String)
3. Monotone
4. Loudness and Amplitude
 - Unit of loudness in decibels

- Theme 8: Electricity

1. Household consumption of electric energy (kilowatt hour)
2. Identify live wire, neutral wire and earth wire in terms of their energy and path they travel
3. Safety Components (fuses/circuit breakers (Qualitative approach only)/grounding)
4. Static Electricity
 - Conservation of charges
 - Conduction
 - Induction
 - Electroscope (Gold Leaf Electroscope)
 - Lightning Conductor
5. Battery as a collection of cells connected in series
6. Dangers of electricity